

Stream Tiered Assessment Framework (STAF)

A structured approach to assessing stream ecosystem condition

Background and Need

USACE planners, engineers, and practitioners need tools for holistically assessing stream structure, function, and dynamic processes. Existing approaches have often relied on species-specific habitat models that focus narrowly on single taxa and do not capture ecosystem-scale outcomes. At the same time, the Civil Works planning process (SMART Planning) imposes different modeling needs as projects progress, from rapid site screening with sparse data (1–3 months) through conceptual design (12–20 months) to final design and post-construction monitoring. There is no standardized, nationally applicable framework that scales assessment effort to project phase while keeping results comparable.

A review of 188 stream assessment methods in the U.S. found that most methods do not evaluate the full range of ecosystem functions, there is limited guidance on matching effort to project phase, and results are difficult to compare across methods, projects, and regions.

The Framework

Screening Tier	Rapid Tier	Detailed Tier
Goal: Provides a broad, watershed or landscape-scale snapshot. Identifies constraints, opportunities, and where additional effort may be needed. Data sources: Desktop GIS, existing data review, limited site photos, and brief recon where available. Effort: Minutes to hours. Uses: Scoping and site screening.	Goal: Refines understanding of site conditions. Field evidence supports planning decisions and alternatives comparison. Data sources: Focused field observations, many site photos, limited modeling or lab data, desktop GIS, existing data. Effort: Hours to days. Uses: Design, comparing alternatives.	Goal: Provides high confidence, site-specific data to support final design decisions and performance. Data sources: Intensive field data collection, extensive modeling or laboratory analysis, site photos, desktop GIS, existing data. Effort: Days to weeks. Uses: Final design, post-construction monitoring, regional studies.

Lower ←————— Effort, data needs, confidence —————→ Higher

Common foundation: Stream functions are consistent across all tiers. The functions are comprehensive and broadly applicable to a wide range of streams.

Comparable results: Outcomes and Ecosystem Condition

Outcomes are observable or quantifiable results that are linked to how one or more stream functions operate.

Physical Outcomes	Chemical Outcomes	Biological Outcomes
Physical outcomes are measurable results of hydrologic, hydraulic, geomorphic, and habitat-forming processes. They describe the structure and how water and sediment move through the system. Examples: floodplain inundation frequency, channel stability, sediment transport balance, habitat unit distribution, substrate composition, and wood abundance.	Chemical outcomes are measurable results of water chemistry and biogeochemical processes. They describe how chemical conditions support or limit aquatic life and ecosystem processes. Examples: dissolved oxygen, nutrient concentrations, temperature regime, pH, contaminant levels, and organic matter decomposition rates.	Biological outcomes are measurable characteristics of aquatic and riparian communities. They describe the presence, abundance, diversity, and functional roles of organisms in the system. Examples: fish assemblage composition, macroinvertebrate diversity, species richness, presence of sensitive taxa, and functional feeding group distribution.

Ecosystem Condition

The overall state of a stream system, based on the combined performance of physical, chemical, and biological outcomes relative to expected or reference conditions. It indicates how well the system sustains ecological function and resilience over time.

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Purpose, Functions, Scoring, and Uses

Purpose and Objectives

STAF provides a nested toolkit of stream assessment methods that align with project planning needs at varying levels of effort. The framework provides a common, transferable structure for stream assessment that can be applied at multiple scales and tailored to local needs.

- Develop a common and transferable structure for stream assessment, applicable at multiple scales and tailored to local needs.
- Compile tools for assessing stream outcomes at three scales: screening-level analyses, rapid field assessment, and detailed analyses informed by empirical data collection.
- Develop select tools to fill gaps in the existing toolbox.
- Demonstrate application of the multi-scale framework as a go-by for users.

Common Stream Functions

- All tiers assess the same set of stream functions.
- The tiers differ in how each function is evaluated and documented.
- This improves consistency across projects and makes it easier to compare results across sites, watersheds, and study phases.

Scoring Approach

- Each stream function is evaluated using tier-appropriate evidence and scored on a 0–15 scale.
- Function scores are grouped into three outcome categories: **physical** (hydrology, hydraulics, geomorphology), **chemical** (water chemistry, biogeochemical processes), and **biological** (aquatic and riparian communities).
- Each outcome category produces a normalized sub-index (0–1). The three sub-indices are averaged to **Ecosystem Condition Index**.
- This structured rollup supports interpretation, comparison across sites, and decision-making.

Typical Uses

- **Ecosystem restoration planning and feasibility studies:** Identify limiting functions, compare reaches, and support alternative development.
- **Project prioritization:** Screening and Rapid tiers support ranking and selection across multiple candidate sites.
- **Planning and regulatory documentation:** Consistent structure for reporting and communicating findings.
- **Design support and monitoring:** Detailed tier supports design decisions and performance evaluation when higher certainty is required.

Resources

- **Website:** usace-wrises.github.io/staf
- **Review of Assessments:** Stepchinski, L. M., McKay, S. K., Harris, A. E., & Menichino, G. T. (2025). [A Review of Stream Assessment Methods in the United States](#). Journal of the American Water Resources Association, 61.
- **Stream functions paper:** Stepchinski, L. M., McKay, S. K., & Menichino, G. T. (In review). [Synthesis and inventory of stream functions](#). Manuscript submitted for publication.
- **Tiered approach paper:** Stepchinski, L. M., McKay, S. K., & Menichino, G. T. (In review). [Tiered Approach to Assessing Stream Ecosystem Condition](#). Manuscript submitted for publication.
- **SFARI Rapid Assessment:** David, G. C., Stepchinski, L. M., Wiest, S. R., & Menichino, G. T. (In review). [Stream Functions Assessment and Rapid Index \(SFARI\): A nationally applicable, rapid, function-based stream assessment](#). ERDC/EMRRP Technical Report. Vicksburg, MS: Army Engineer Research and Development Center.

STAF provides a clear tiered pathway, a consistent set of stream functions, and a structured scoring approach that rolls function results into physical, chemical, and biological outcomes and an overall Ecosystem Condition Index. It helps teams scale effort to need, keep evaluations comparable, and communicate results clearly.